Practical Assignment 1.1: Implementation of Intelligent Agent : Simple Reflex Agent

Name: Vedangi Patil

Batch: A1

Roll No: 19

PRN: 202401040058

Title:-

Theatre/Auditorium Cleaning using a Simple Reflex Vacuum Agent

Aim:- To design and implement a Simple Reflex Vacuum Agent that cleans different types of waste and dirt in a theatre based on the current condition of the floor.

Objectives

1. To understand the working of a Simple Reflex Agent and how it reacts to the current state of an environment.
2. To model a theatre environment with different floor conditions such as clean, dusty, popcorn waste, and drink spills.
3. To implement different cleaning actions such as dust cleaning, waste collection, and spill cleaning according to the detected condition.
4. To evaluate the agent's performance using a simple score based on cleaning actions and movement between locations.

Conceptual Details A Simple Reflex Agent is an intelligent agent that decides what to do based only on the current condition of the environment. It does not remember previous conditions or try to predict what will happen next.

In this practical, the environment is a theatre/auditorium with two locations, A and B. Each location can randomly have one of four conditions:

- Clean – no cleaning is required.
- Dusty – the agent performs dust cleaning.
- Popcorn Waste – the agent activates waste collection.
- Drink Spill - the agent performs spill cleaning.

The vacuum starts at a randomly selected location and checks its current condition. Based on that condition, it immediately performs the appropriate action. After cleaning the first location, it moves to the other location and performs the same process.

The performance of the agent is measured using a score. Cleaning gives positive points, while movement between locations reduces the score slightly.

The basic working can be represented as:

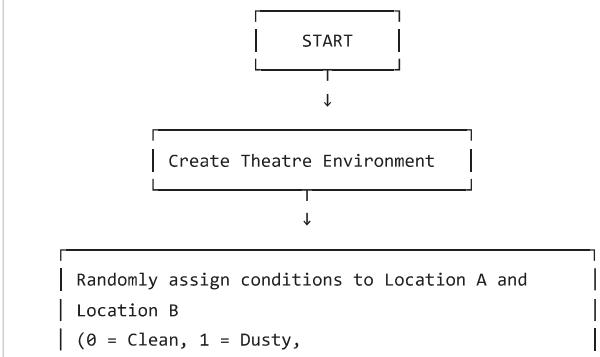
Current Condition → Appropriate Action → Location Becomes Clean

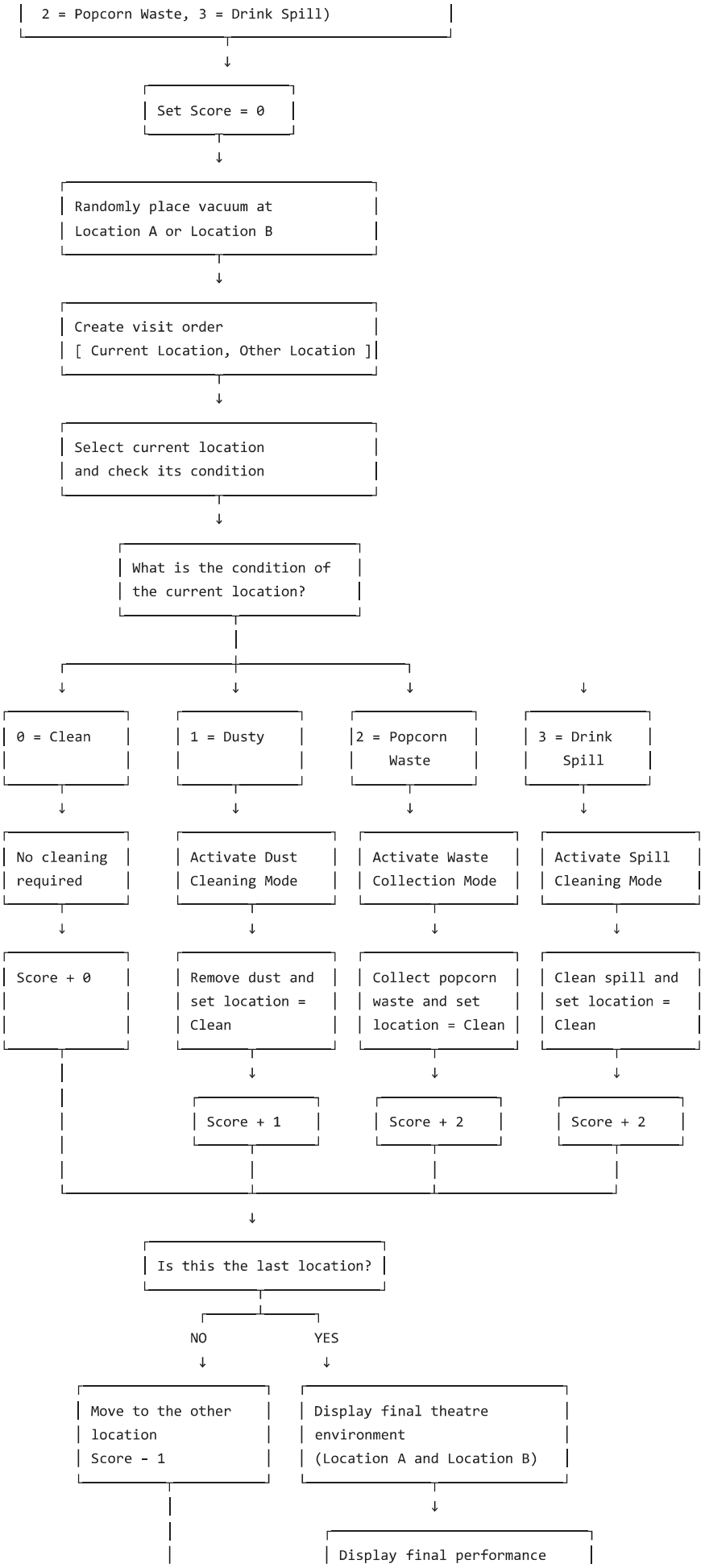
For example:

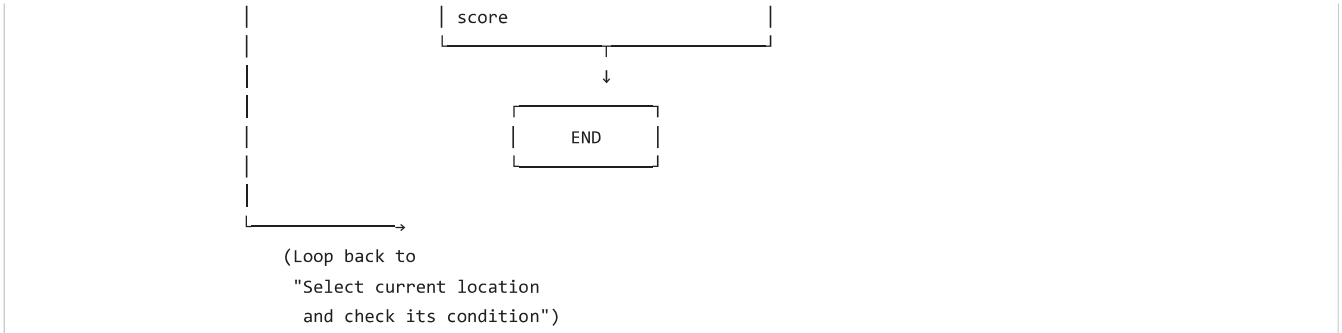
Popcorn Waste → Waste Collection → Clean

This makes the system a simple reflex agent because its action depends only on what it currently senses.

Flowchart




2 = Popcorn Waste, 3 = Drink Spill)
Set Score = 0
Randomly place vacuum at Location A or Location B
Create visit order [Current Location, Other Location]
Select current location and check its condition
What is the condition of the current location?
0 = Clean
1 = Dusty
2 = Popcorn Waste
3 = Drink Spill
No cleaning required
Activate Dust Cleaning Mode
Activate Waste Collection Mode
Activate Spill Cleaning Mode
Score + 0
Remove dust and set location = Clean
Collect popcorn waste and set location = Clean
Clean spill and set location = Clean
Score + 1
Score + 2
Score + 2
Is this the last location?
NO
YES
Move to the other location
Score - 1
Display final theatre environment (Location A and Location B)
Display final performance



Algorithm Step 1: Start the program.

Step 2: Create the theatre environment with two locations, A and B.

Step 3: Assign a random condition to each location from the following:

Clean Dusty Popcorn Waste Drink Spill

Step 4: Randomly place the vacuum cleaner at either Location A or Location B.

Step 5: Check the current condition of the location.

Step 6: If the location is Clean, take no cleaning action.

Step 7: If the location is Dusty, activate the dust cleaning action and make the location clean.

Step 8: If the location contains Popcorn Waste, activate waste collection and make the location clean.

Step 9: If the location has a Drink Spill, activate spill cleaning and make the location clean.

Step 10: Update the performance score according to the action performed.

Step 11: Move the vacuum to the other location and reduce the score for movement.

Step 12: Check the condition of the second location and perform the appropriate cleaning action.

Step 13: Display the final condition of both theatre locations.

Step 14: Display the final performance score.

Step 15: Stop the program.

Conclusion:

The Theatre Vacuum Agent successfully cleans both theatre locations based on their floor conditions. It uses a simple reflex approach by taking actions according to the current condition of each location. Different cleaning modes are activated for dust, popcorn waste, and drink spills. The agent updates each cleaned location to Clean after performing the required action. A performance score is calculated based on cleaning performed and movement between locations. Thus, the system demonstrates how a simple reflex agent can efficiently maintain a clean theatre environment.

```
import random

class Environment:

    def __init__(self):

        # Theatre floor conditions
        # 0 = Clean
        # 1 = Dusty
        # 2 = Popcorn Waste
        # 3 = Drink Spill

        self.locationCondition = {
            'A': random.randint(0, 3),
            'B': random.randint(0, 3)
        }


class TheatreVacuumAgent:

    def __init__(self, environment):

        print("Initial Theatre Environment:")
```

```
        print(environment.locationCondition)

        # Performance measurement
        score = 0

        # Randomly place vacuum at A or B
        currentLocation = random.choice(['A', 'B'])

        print("\nVacuum starts at Location", currentLocation)

        # Visit both theatre sections
        locations = [currentLocation]

        if currentLocation == 'A':
            locations.append('B')
        else:
            locations.append('A')

        # Clean both locations
        for location in locations:

            condition = environment.locationCondition[location]

            print("\nChecking Location", location)

            if condition == 0:
                print("Location", location, "is Clean.")
                print("No cleaning required.")

            elif condition == 1:
                print("Location", location, "is Dusty.")
                print("Activating Dust Cleaning Mode...")

                environment.locationCondition[location] = 0
                score += 1

                print("Dust has been removed.")

            elif condition == 2:
                print("Location", location, "has Popcorn Waste.")
                print("Activating Waste Collection Mode...")

                environment.locationCondition[location] = 0
                score += 2

                print("Popcorn waste has been collected.")

            elif condition == 3:
                print("Location", location, "has a Drink Spill.")
                print("Activating Spill Cleaning Mode...")

                environment.locationCondition[location] = 0
                score += 2

                print("Drink spill has been cleaned.")

            # Move to next location
            if location != locations[-1]:
                print("Moving to Location", locations[1])

                score -= 1

        print("\nFinal Theatre Environment:")
        print(environment.locationCondition)

        print("\nPerformance Measurement:", score)


# Create the theatre environment
theEnvironment = Environment()

# Create the simple reflex vacuum agent
theVacuum = TheatreVacuumAgent(theEnvironment)
```

```
Initial Theatre Environment:
{'A': 1, 'B': 1}
```

```
Vacuum starts at Location B

Checking Location B
Location B is Dusty.
Activating Dust Cleaning Mode...
Dust has been removed.
Moving to Location A

Checking Location A
Location A is Dusty.
Activating Dust Cleaning Mode...
Dust has been removed.

Final Theatre Environment:
{'A': 0, 'B': 0}

Performance Measurement: 1
```